

Education

- University of Waterloo — B.A.Sc. Electrical and Computer Engineering (Term 3B)

Technical Skills

Languages: C, C++, Python, ARM assembly, Bash, VHDL, SQL, MATLAB

Libraries: Pandas, os, Plotly, NumPy, pthreads, CMock, STL, OpenCV, TensorFlow, Keras

Environments and tools: Git, Linux, Jenkins, Jira, DuckDB, Nios II, STM32, Artifactory, VS Code

Experiences

Statistics Canada - Junior Developer, EQ Innovations & Automation - Ottawa, Canada

Jan 2026 – Apr 2026

- Developed a generalized Python data ingestion pipeline to convert fixed width datasets with varying schemas and encodings into Parquet, integrating DuckDB and pandas for efficient downstream processing.
- Designed a custom fixed width ingestion algorithm that parsed records using configurable widths and column specifications, handled byte and character-based encodings, validated malformed records, and staged normalized data for Parquet export.
- Built a reusable R development repository template and CI/CD pipeline for VS Code, automating code formatting, validation/error checks, compatibility checks, package builds, and publishing to Artifactory.

Ford - Firmware Developer - Kanata, Canada

Sept 2023 – Dec 2023

- Automated Polyspace static analysis workflows with Python integration for Jenkins and Excel, cutting daily build runtime through job chaining.
- Authored and refactored embedded C unit tests for FNV4 power management firmware, improving code coverage from 24% to 91% through Unity, CMock, and Polyspace.

ECCC - Scientific Programmer – Ottawa, Canada

Jan 2023 – May 2023

- Built Python tools using NumPy and RMN to calculate wind and pressure fields from satellite model outputs.
- Increased model stability and maintainability through systematic debugging and deployment of a GitLab issue tracking system, shortening debugging cycles.

Projects

Real Time Executive (RTOS Kernel)

May 2026 – Aug 2026

- Developed a real-time executive in C for an STM32 ARM Cortex M4, implementing dynamic task creation and termination, SVC based kernel operations, and PendSV context switching for concurrent task execution.
- Built a custom first fit dynamic memory allocator with free list management, 4-byte alignment, per task memory ownership, block coalescing, and external-fragmentation tracking without using malloc().
- Implemented a SysTick driven preemptive Earliest Deadline First (EDF) scheduler with dynamically allocated task stacks, deadline-based priorities, task periodicity, and runtime deadline updates.

Ottohack8 focus monitor

Jan 2026 – Jan 2026

- Built a real time focus monitoring prototype using Solace Agent Mesh to analyze camera derived physiological and behavioral data and trigger focus alerts.
- Developed a Python UDP pipeline to parse and normalize live sensor data, including heart rate, breathing, blink, and speech events, into JSON for agent processing.

Quantitative trading Algorithm

Jan 2026 – present

- Developing and backtesting an ETF trading strategy combining trend following and mean reversion models with volatility weighted allocation, backtested performance showing up to 70% higher cumulative returns than SPY.
- Built a Python pipeline for data ingestion and feature engineering using NumPy and Pandas with market data retrieved via yfinance and Bloomberg.

Graph Based Route Optimization Program

May 2024 – Jun 2024

- Built an interactive C++ traffic-network application using nested hash maps and priority queues to support dynamic graph updates and path queries.
- Implemented Dijkstra's shortest path algorithm with dynamic edge weighting, reducing average query time compared with a naive search approach.

Interrupt Driven FPGA Stimulus-Response System

Sept 2025 – Oct 2025

- Built a real time polling and interrupt system on an Altera FPGA using NIOS II and Avalon bus.
- Benchmarked polling vs interrupt designs, achieving >40 % CPU efficiency gain and sub millisecond latency.